PRIYANKA BOLEM

AI AND MACHINE LEARNING ENGINEER
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SUMMARY

AI and ML engineer with 6+ years in software engineering, 5+ of them shipping production ML and AI systems for healthcare, enterprise and banking clients: an LLM-based CSR assist that cut average handle time about 30 percent in a 2-million-call member contact center, chatbots and voicebots on Microsoft Foundry, an AI-driven sales and service application over an internal system with millions of records, and predictive models deployed on AWS SageMaker with drift monitoring. Strong in Python, SQL, PyTorch, TensorFlow, intent classification, RAG, and MLOps on AWS and Azure. MS in Computer Science.

TECHNICAL SKILLS

Languages: Python, SQL, Java, R

ML and Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, Random Forest, Hugging Face Transformers, OpenCV, NLP, Vector Databases

Modeling and Optimization: Regression, Classification, Clustering, Neural Networks, Bayesian Optimization, Monte Carlo Simulation, Genetic Algorithms, Reinforcement Learning, Hyperparameter Tuning, Model Calibration, Uncertainty Estimation

Data and Pipelines: Apache Spark, Kafka, Airflow, Azure Databricks, Redis, Microsoft Dynamics CRM, ETL design, Feature Engineering, DVC

MLOps and Deployment: AWS SageMaker (Pipelines, Endpoints), MLflow, Docker, Kubernetes, FastAPI, GitHub Actions, CI/CD, CloudWatch, Evidently, Prometheus, Grafana, Drift Detection

Cloud: AWS (SageMaker, Lambda, EC2, ECS, S3), Azure (ML Studio, Databricks), Microsoft Foundry, GCP (Vertex AI, BigQuery, Cloud Run)

Evaluation and Experimentation: A/B Testing, Cross-validation, ROC-AUC, Precision/Recall/F1, AUPRC, Model Calibration

Generative AI and NLP: GPT-4, BERT, FinBERT, LLaMA, LangChain, LangGraph, RAG, Vertex AI Search, Microsoft Foundry, Portkey, Intent Classification, Summarization, Chatbots, Voicebots, Genesys, Prompt Engineering, Tokenization, Neo4j

Visualization: Tableau, Power BI, Matplotlib, Seaborn, Plotly, Pandas, NumPy, Statsmodels

PROFESSIONAL EXPERIENCE

AI Engineer | Premera Blue Cross (via Adisys) *Mountlake Terrace, WA | Jul 2026 - Present*

- Leading the AI automation of Premera's member call center on Microsoft Dynamics CRM across voice, chat and email, about 2 million calls and 100,000 chats and emails a year across roughly 1,000 representatives.
- Built an intent classification model on historical member inquiries and use the predicted intents to route and streamline handling across all three channels.
- Built the CSR assist, which during a call pulls and summarizes the member's data from the claims, billing and member systems so the representative answers without switching applications; average handle time fell from about 13 minutes to about 9, roughly 30 percent.
- Automated the post-call documentation, generating the interaction summary and creating the inquiry record that representatives used to write by hand, raising summary quality and consistency; the captured data feeds reporting and future models.
- Building the member portal chatbot and the IVR voicebot through the Genesys APIs, both driven by the intents model, to answer common inquiries and escalate to a representative on request (Portkey to Microsoft Foundry, Redis, Dynamics CRM and AWS).

Machine Learning Engineer | HP Inc. (via Adisys) *Remote | Dec 2025 - Jun 2026*

- Built the sales and service application for Fiji, HP's internal sales and service system, connecting its data through APIs to Microsoft Foundry so that both the sales lifecycle and the service lifecycle run on AI-driven decisions applied over HP's business rules.

- Embedded LinkedIn data into Dataverse with AI recommendations so sales representatives get account context without the manual lookups that had slowed the sales process, and close deals faster.
- Analyzed millions of Fiji records on how projects progressed and how factories received them, using AI and ML to find where the project lifecycle could be improved.
- Built dashboards presenting the AI analytics on sales and operations to leadership.

Machine Learning Engineer | Truveta (via Adisys) *Seattle, WA | Feb 2025 - Nov 2025*

- Deployed end-to-end ML pipelines on AWS SageMaker, using SageMaker Pipelines for training automation, SageMaker Endpoints for real-time risk scoring, and CloudWatch for production monitoring and alerting, built to healthcare data compliance standards.
- Designed and implemented predictive models in Python and TensorFlow to forecast chronic disease progression from de-identified EHR and demographic data, enabling automated patient stratification for clinical research and population health studies.
- Applied Bayesian Optimization and Monte Carlo simulation across large patient cohorts to quantify uncertainty in outcome predictions, improving prediction precision and providing data-driven insight into intervention effectiveness.
- Built a hybrid optimization framework pairing Genetic Algorithms for cohort selection with Reinforcement Learning for adapting longitudinal strategies, integrating historical records with real-time signals for continuous learning; improved plan efficiency and outcome prediction accuracy by 22%.

Machine Learning Engineer | Capgemini *India | Apr 2019 - Nov 2022*

- Built and deployed models with Scikit-learn, XGBoost, and LightGBM to predict credit default probability, loan delinquency, and fraudulent claims, then embedded them into origination, collections, and fraud triage systems with risk and compliance partners, reducing end-to-end loan decision cycle time by 25%.
- Engineered features from large-scale financial, credit bureau, and behavioral data using Pandas and NumPy, capturing payment patterns, credit utilization, delinquency history, and spending trends.
- Built automated ETL pipelines in Python and SQL across core banking systems, credit bureaus, and CRM platforms, reducing data preparation time by 40%; scaled preprocessing, feature engineering, and distributed training on Azure Databricks.
- Orchestrated Apache Airflow workflows for scheduled retraining, data refresh, and batch inference, keeping predictions accurate as customer behavior and regulation changed.

Junior Software Engineer | Halcyon Technologies *India | Feb 2018 - Mar 2019*

- Trained in Java, Python and Android development, then built and maintained Android mobile applications in Java, implementing features and fixing defects under senior engineers' review.

PROJECTS

- **Financial Sentiment MLOps Platform.** End-to-end NLP platform: FinBERT fine-tuning, MLflow tracking, FastAPI + Docker + Kubernetes serving, Prometheus/Grafana monitoring, Evidently drift detection, GitHub Actions CI/CD; live on Hugging Face Spaces. github.com/priyankabolem/financial-sentiment-mlops • live demo
- **Grocery Product Search and Re-ranking on Google Cloud.** Two-stage search over an Instacart catalog in BigQuery: Vertex AI Search retrieval, a gradient boosting re-ranker on a Vertex AI endpoint scoring reorder propensity, and a FastAPI service on Cloud Run blending both into the final ranking with a retrieval-order fallback; Docker, tests, CI. github.com/priyankabolem/grocery-search-vertex
- **Credit Card Fraud Detection: Production MLOps.** Fraud scoring on a 1:578-imbalanced dataset, chronological split: XGBoost vs logistic baseline, test AUPRC 0.765 vs 0.718 and ROC-AUC 0.979, threshold set for a 0.90 precision budget, FastAPI serving, PSI drift monitoring, Docker, CI with 10 tests. github.com/priyankabolem/fraud-detection-mlops

EDUCATION

Master of Science, Computer Science | Northwest Missouri State University, Maryville, MO *Jan 2023 - Dec 2024*